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Roll No

EC-221-CBCS

B.E. III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Digital Circuits and Systems

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Design Ex-OR and Ex-Nor gate using
i) NAND gate ii) NOR gate
b) Reduce the following using Karnaugh map technique and realize using basic gates
i) $F(A, B, C, D) = \sum_m(5, 6, 7, 12, 13) + \sum_d(4, 9, 14, 15,)$
ii) $F(A, B, C, D) = \sum_m(2, 4, 8, 11, 15) + \sum_d(1, 10, 12, 13,)$
2. a) Design full adder using two half adders.
b) Design BCD adder.
3. Minimize the expression using Quine McCluskey method
$$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}D$$
4. a) Design a carry look ahead adder.
b) Design a magnitude comparator

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5. Explain the designing and working of following flip flop.
 - i) SR flip flop
 - ii) D flip flop
 - iii) T flip flop.
6. Explain the working of JK flip flop. What is race around condition? How can it be overcome?
7.
 - a) Explain the working of asynchronous Up/Down counter.
 - b) Discuss about different types of register's and their application.
8. Write short notes on any two of the following:
 - a) DTL, TTL
 - b) CMOS inverter
 - c) ROM, RAM
